Phase 2 – Data Collection & Preparation

Objectives

The goal of this phase is to gather the required data for your project and transform it into a clean, usable format ready for analysis.

Tasks

1. **Data Collection**
    - Gather data from the chosen sources (internal datasets, open data portals, APIs, sensors, web scraping, etc.).
    - Document each source clearly (URL, access method, dataset description, retrieval date).
2. **Data Preprocessing**

    - **Data Acquisition Validation**
        - Confirm dataset integrity (readable, correct format, no corrupted records)
    - **Missing Data Handling**
        - Identify missing values
        - Apply a reasonable handling method (remove or impute)
    - **Duplicate Removal**
        - Detect and remove duplicate rows or records
    - **Outlier Detection & Treatment**
        - Identify unusual values and decide whether to keep, fix, or remove them
    - **Data Type Correction**
        - Ensure all features have correct data types (numeric, categorical, dates, etc.)
    - **Formatting & Standardization**
        - Clean up inconsistent formatting (capitalization, date formats, units, etc.)
    - **Categorical Data Encoding**
        - Convert categorical features into numerical formats suitable for analysis
    - **Feature Scaling / Normalization**
        - Apply appropriate scaling if numeric features are on different ranges
    - **Data Integration** *(if multiple sources used)*
        - Merge datasets and handle alignment inconsistencies

3. **Documentation**
    - Record any problems you encountered (e.g., limited access, inconsistent formats).
    - Provide a detailed list of preprocessing steps, including justification for any decisions made.

Deliverables

A packaged dataset including:

- **Raw data** (as originally collected)
- **Cleaned/processed dataset**
- **Source Code**
- **Preprocessing documentation** (1–2 pages or well-structured README), including:
    - Data source descriptions
    - Cleaning and transformation steps
    - Summary of challenges and solutions